

SL Math School: Random Growth Models, etc.

Combined Tutorial Problems

1 Brownian Motion I

1.1 Prologue

1.1.1 Dyadic Construction of Brownian Motion

For $n \in \mathbb{N}$ and $k \in \{1, 3, \dots, 2^n - 1\}$ or $(n, k) = (0, 1)$, define an independent family of $\mathcal{N}(0, 1)$ distributed random variables $\Gamma_{n,k}$. Recall that we defined Brownian Motion to be the uniform limit of the functions B_n , defined so that $B_0(x) = x\Gamma_{0,1}$ and

$$B_{n+1}\left(\frac{k}{2^n}\right) = \frac{1}{2} \left(B_n\left(\frac{k-1}{2^n}\right) + B_n\left(\frac{k+1}{2^n}\right) \right) + 2^{-(n+1)/2} \Gamma_{n,k},$$

and B_{n+1} linearly interpolates between its values on $[0, 1] \cap 2^{-n}\mathbb{Z}$. For (n, k) , define the family of σ -algebras $\{\mathcal{F}_{n,k}\}$ as follows

$$\mathcal{F}_{n,k} = \sigma \{ \Gamma_{m,j} \mid (m, j) \neq (n, k) \},$$

Define the corresponding conditional probability to be

$$\mathcal{P}_{n,k}(A) = \mathbb{E}[\mathbb{1}_A \mid \mathcal{F}_{n,k}],$$

for any event A . Notice that $\mathcal{P}_{n,k}(A)$ is therefore a $\mathcal{F}_{n,k}$ -measurable random variable.

We also refer to a Brownian Bridge on $[0, 1]$ conditioned to end at a to be a Brownian Motion conditioned so that $B(1) = a$.

For this first tutorial, we will work on a probability space $\mathbb{P}$ where the $\Gamma_{n,k}$ are all defined and independent, and conditioned on the almost sure event where B_n converges uniformly to the function B .

1.2 Tutorial 1: Problems

1.2.1 Warm-Up Problems

Problem 1.

Fix $a \in \mathbb{R}$. Show that

$$\{B(1) = a\} = \{\Gamma_{0,1} = a\}.$$

Find the analogous equality for more general (n, k) . That is, for $n \geq 1$ and $k = 1, 3, \dots, 2^{n-1}$,

$$\{\Gamma_{n,k} = a\} = \{f_{n,k}(B) = a\},$$

where $f_{n,k} : \mathcal{C}([0, 1]) \rightarrow \mathbb{R}$ is some functional depending on n, k . Find $f_{n,k}$.

Problem 2 (Concentration and Anti-concentration).

Let μ and ν be two (probability) measures so that μ is absolutely continuous with respect to ν and has a Radon-Nikodym derivative

$$f(x) = \frac{d\mu}{d\nu}(x).$$

Show that for all measurable sets A

$$\left(\operatorname{ess\,inf}_{x \in A} f(x) \right) \nu(A) \leq \mu(A) \leq \left(\operatorname{ess\,sup}_{x \in A} f(x) \right) \nu(A).$$

where

$$\begin{aligned} \operatorname{ess\,sup}_A f &= \inf\{M \in \mathbb{R} : f \leq M \text{ for } \nu\text{-a.e. } x \in A\}, \\ \operatorname{ess\,inf}_A f &= \sup\{m \in \mathbb{R} : f \geq m \text{ for } \nu\text{-a.e. } x \in A\}. \end{aligned}$$

Remark: We refer to the first inequality as a concentration inequality and the second as an anti-concentration inequality. Do you see why?

1.2.2 Core Problems

Problem 3 (Stochastic Domination).

Let X and Y be real-valued random variables. Suppose that for every $x \in \mathbb{R}$,

$$\mathbb{P}(X \geq x) \geq \mathbb{P}(Y \geq x).$$

In this case, we say that X stochastically dominates Y .

Show that there exists a probability space $(\Omega, \mathbb{P})$ carrying random variables $\tilde{X}$ and $\tilde{Y}$ such that

$$\tilde{X} \stackrel{d}{=} X, \quad \tilde{Y} \stackrel{d}{=} Y, \quad \mathbb{P}(\tilde{X} \geq \tilde{Y}) = 1.$$

A picture proof is enough. Do this in the following cases:

- (a) X and Y both have density functions.
- (b) X and Y are both discrete random variables.
- (c) Convince yourself that the same statement is true for arbitrary real-valued random variables.

Problem 4 (Monotonic Coupling of Brownian Bridges).

Fix real numbers a_1, a_2, b_1, b_2 so that $a_1 < a_2$ and $b_1 < b_2$. Let B_1 and B_2 be Brownian Bridges on $[0, 1]$ so that $B_i(0) = a_i$ and $B_i(1) = b_i$ for $i = 1, 2$. Show that there exists probability space $(\Omega, \mathbb{P})$ with two random-continuous functions $\tilde{B}_1, \tilde{B}_2$ so that

$$\tilde{B}_1 \stackrel{d}{=} B_1, \quad \tilde{B}_2 \stackrel{d}{=} B_2, \quad \mathbb{P}(B_1(x) \leq B_2(x) \text{ for all } x \in [0, 1]) = 1.$$

Problem 5. (a) Show that there is a $\mathcal{F}_{1,1}$ -measurable random function $\tilde{B}$ so that for all $0 \leq x \leq 1$,

$$B(x) = \phi(x)\Gamma_{1,1} + \tilde{B}(x).$$

where $\phi(x)$ is the following piece-wise linear function:

$$\phi(x) = \begin{cases} x & 0 \leq x \leq \frac{1}{2} \\ 1 - x & \frac{1}{2} \leq x \leq 1. \end{cases}$$

- (b) Let $I \subset (0, 1)$ be a closed interval. Show there exists a $\mathcal{F}_{1,1}$ -measurable random function $Y = Y(M)$ defined for $M > 0$ so that

$$\left\{ \max_{x \in I} B(x) = M \right\} = \{\Gamma_{1,1} = Y(M)\}.$$

Define Y in terms of its inverse function.

(c) Prove for any $M \geq 0$ that

$$\mathbb{P} \left(M - \frac{\epsilon}{2} \leq \max_{x \in [\frac{1}{4}, \frac{3}{4}]} B(x) \leq M + \frac{\epsilon}{2} \right) \leq \frac{4}{\sqrt{2\pi}} \epsilon.$$

Conclude that $\max_{x \in [\frac{1}{4}, \frac{1}{2}]} B(x)$ has a continuous distribution function (i.e. it has no atoms).

(d) Let B_t now be a Brownian Motion defined for all $t \geq 0$. Show that for all $M_1, \dots, M_n \geq 0$,

$$\mathbb{P} \left(\max_{i=1, \dots, k} \left| \max_{t \in i + [\frac{1}{4}, \frac{3}{4}]} B_t - M_i \right| \leq \epsilon \right) \leq \left(\frac{4}{\sqrt{2\pi}} \epsilon \right)^k$$

Hint: For part (b), consider the function

$$S(z) = \max_{x \in I} z\phi(x) + \tilde{B}(x).$$

Show that this function is one-to-one and onto. In particular, in the case $I = [\frac{1}{4}, \frac{3}{4}]$, for $z_1 < z_2 \in I$

$$S(z_2) - S(z_1) \geq \frac{1}{4}(z_2 - z_1).$$

1.2.3 Extra Challenge Problems

Problem 6 (Modulus of Continuity for Brownian Motion).

Define the following modulus of continuity constant for Brownian Motion on $[0, 1]$.

$$C = \sup_{0 < \delta < \frac{1}{2}} \sup_{0 < |s-t| < \delta} \frac{|B(s) - B(t)|}{\sqrt{\delta \log(1/\delta)}}. \quad (1)$$

In this question, we will show that there exists constants $c, d > 0$ so that

$$\mathbb{P}(C \geq m) \leq ce^{-dm^2}. \quad (2)$$

Define

$$\Delta_n = \max_{0 \leq i < 2^n} |B(\frac{i+1}{2^n}) - B(\frac{i}{2^n})|,$$

$$D = \sup_{n \geq 1} \frac{\Delta_n}{\sqrt{n} 2^{-n/2}}.$$

(a) Show that for all $|s - t| \leq 2^{-n}$

$$|B(s) - B(t)| \leq 2 \sum_{m=n}^{\infty} \Delta_m$$

(b) Show that for $a > 0$ large enough, there exists constants $c, d > 0$ so that

$$\mathbb{P}(D > a) \leq ce^{-da^2}.$$

(c) Show on the event $\{D \leq a\}$ that

$$\sup_{s, t \in [0, 1], |s-t| \leq 2^{-n}} |B(s) - B(t)| \leq 2a \sum_{m=n}^{\infty} \sqrt{m} 2^{-m/2}.$$

(d) By first considering the case for dyadic $\delta = 2^{-n}$, prove (2).